Henok Demissie

Software Engineer/ ML Engineer

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EDUCATION

B.Sc. Software Engineering *05/2022 – 06/2026*

Adama Science and Technology University | CGPA: 3.82 / 4.0 *Adama, Ethiopia*

- Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Information Retrieval, Artificial Intelligence, Computer Networks, Computer Architecture, Operating Systems, Concurrency, Object-Oriented Design, Advanced Java, C++ Programming, Requirements Engineering.

Real-Time Backend Systems Program *07/2025 – 05/2026*

EPIC Institute of Technology *Newtown, Pennsylvania*

- Designed and implemented real-time backend architectures using distributed systems, Raft consensus, and fault-tolerant design.
- Built large-scale data pipelines with AWS and Docker; applied time series analysis for real-time insights.

Competitive Programming and Software Development *11/2023 – 09/2024*

Africa To Silicon Valley *Adama, Ethiopia*

- Solved 500+ algorithmic problems on LeetCode and Codeforces.

PROFESSIONAL EXPERIENCE

Robotic Games Inc. (Funded by Founders) *08/2025 – Present*

Software Engineer *San Francisco, CA*

- Applied SimHash-based deduplication with adaptive Hamming thresholds, reducing redundant analyses by 60-80% using Python and Redis.
- Created a content-matching pipeline on AWS (FastAPI, PostgreSQL), lowering API costs by 60% and reducing median latency from 2.3 s to 0.8 s through Dockerized services and optimized indexing.
- Developed a browser extension (React, WebExtensions API) that filters unwanted web content in real-time, cutting scan time to under 20 ms.
- Engineered a low-latency backend (< 370 ms) with FastAPI and Redis, integrating an on-device privacy layer for secure personalization.

Africa to Silicon Valley *01/2025 – Present*

Head of Education *Adama, Ethiopia*

- Mentored 60+ students in software development and data structures; organized 19 contests and authored solutions, resulting in 22 internships and job placements.
- Recognized as Best Lecturer among eight educators for instructional quality and student engagement.

Eskalate *12/2024 – 08/2025*

Project Lead (Sparck, AI Social Media Platform) *Palo Alto, CA*

- Led a 15-member team to develop Sparck, a social media management platform for creation, scheduling, multichannel publishing, and analytics with 98.9% uptime.
- Created 50+ RESTful APIs with OAuth2/JWT authentication, rate limiting, and cursor pagination; improved observability with structured logs, traces, and SLOs.
- Reduced feature lead time by 30% via query optimization, Redis caching, and connection tuning.
- Containerized services with Docker and automated CI/CD pipelines using GitHub Actions.

Software Engineering Intern *07/2024 – 09/2024*

Eskalate *Palo Alto, CA*

- Developed socket-based chat features supporting 500+ concurrent users with zero-downtime releases.
- Improved code quality by driving prototyping, testing, and reviews, increasing test coverage from 55% to 78%.

SKILLS

Languages: Python, C, C++, Java, JavaScript, Dart, Rust, SQL

Frameworks & Libraries: Django REST Framework, FastAPI, Next.js, React, Flutter, Flask, Express.js

ML/AI: PyTorch, scikit-learn, Hugging Face Transformers, RAG, Vector Search (FAISS, Pinecone), NumPy, Pandas

Databases: MySQL, MongoDB, PostgreSQL, SQLite

Cloud / Tools: AWS, GCP, Docker, Git & GitHub, CI/CD, Linux

PROJECTS

Career Pilot — AI-Powered Career Guidance Platform

- Designed a responsive frontend with Next.js and Framer Motion; integrated FastAPI microservices to serve 280+ monthly active users.
- Established a robust authentication system using OAuth 2.0 and role-based access control, resulting in zero incidents of unauthorized access and meeting stringent compliance requirements.

QuizMe – Adaptive AI-Driven Quiz Platform

- Engineered a system with RAG pipelines, vector search, and prompt orchestration, processing over 80 quizzes per day.
- Delivered a prototype using Python, Django, LangChain, and FAISS/Pinecone; ranked Top 30 among 1,000+ teams nationally.

ACHIEVEMENTS AND AWARDS

- African and Arab Collegiate Programming Championship (ACPC) Finalist | Expert Rating on Codeforces Top 5% globally | A2SV Generative AI for Africa Hackathon Semi-Finalist